# Observer-Screen Synchronization as a Candidate Inflation-Free Branch: Record-Density Expansion, Required Premises and a Conditional Tilt

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**Abstract**

For a supplied layered event population whose comoving read law generates the same finite order for every scale-factor profile, the expansion history is a record density per comoving cell and conformal tick, and the redshift between two epochs is the fourth root of a record-count ratio for congruent comoving windows with constant scale factor within each, or in the shrinking-window limit of a continuous profile. These statements are exact and machine-checked; native repair selects neither the population nor its profile. On this footing we state observer-screen synchronization as a candidate inflation-free branch of Observer-Patch Holography (OPH) and list the premises it must supply; none of its replacements for the jobs of inflation is supplied here. A conditional source theorem fixes a screen amplitude from a primitive collar release law and a spectral tilt $n_s = 1 - P_\star/48$ from an edge-center generator once its full-collar generator density is supplied; physical source dilation and radial cross-covariance tomography give two sufficient routes to a unique three-dimensional spectrum, and vanishing spatial curvature holonomy identifies a flat FLRW branch without selecting it. The generator density is a declared branch input, so the tilt is a declared map evaluated at a target-informed pixel. At the comparison pixel located from the measured fine-structure constant the tilt is 0.96602, $0.27\sigma$ above Planck 2018 and at the lower 0.612 percent position of the joint posterior of ACT DR6 with Planck, lensing and DESI 2024 baryon acoustic oscillations, a retrospective tension on data that precede the registration.

**Interpretation and state variables.** OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a

unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

**Basis, premise labels and paper set.** OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

**Time conventions.** Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

**One theory and its executable hypothesis.** OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Coherence, protected

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/docs/BASIS.md

codes, radial directions, fields, clocks and couplings may be specified as part of its design; their implementation and observable consequences must be established on that process. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

**The tower.** The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1+3$ scalar description. Complete finite source factors join spatial CAR dynamics, particle creation, repair and a common calibrated clock. Physical validation of that clock and energy normalization, and an interacting chiral $3+1$ limit, remain open.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. A specified coherent ordering memory protects native exchange-history sectors, and a colored extension implements their fermionic exchange. A localized physical defect, its maintenance-to-mass law and a predicted measured spectrum remain unestablished; the earlier double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

**Completion and stopping.** The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

**Fixed physical boundaries.** Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a specified and implemented clock calibration relates it to duration. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

## This paper's hypothesis and place in the tower

This paper sits at layer seven and develops the primordial branch without an inflaton. Its hypothesis is that observer-screen synchronization does the logical jobs of inflation: expansion is a record density on the sourced causal order, spatial flatness is zero clock-slice curvature holonomy, horizon-scale coherence is same-boundary repair, the nearly scale-invariant spectrum is a geometric screen spectrum with edge-center tilt $n_s = 1 - P_\star/48$, the hot start is a MaxEnt release, and the acoustic sky follows from ordinary Boltzmann transfer.

The unconditional finite results are the record-density form of a spatially flat expanding background on a supplied layered population, the finite quotient ensemble theorems, and the cocycle freedom that leaves the tilt exponent unselected. The screen amplitude, the tilt, the two radial uniqueness routes and the flatness identification are conditional theorems: each holds once its displayed premise is supplied, and no finite source supplies the full-collar generator density $P_\star/24$, the primitive collar release law, the record-production profile or a horizon-coherence mechanism. The FLRW chart, the population and its profile are premises.

For the branch to apply, the executable model must emit from one native process the geometric collar-volume field, the generator density and orientation-half identity, a physical clock calibrating the release, a radial observation algebra that lifts the two-dimensional screen to a three-dimensional curvature field, and the record-production law. The native construction constrains this join. Its continuum generator is not identified with the round Laplacian, its reference state identifies no quantum vacuum, and under a bounded form-congruent identification its spatial covariance forces $\theta = 0$, so the tilt needs an ingredient beyond the native density field (see the paragraph on the native source after the screen-spectrum theorems). The branch is refuted if the frozen tilt protocol excludes $n_s = 0.96602$ on its eligible data, if nonzero running or primordial isocurvature is detected beyond the measurement errors, or if no admissible native ingredient can supply a nonzero tilt.

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/docs/COMPLETION_PROGRAM.md

# 1 Finite quotient theorem scope

**Finite quotient ensemble theorem.** Fix a finite regulator $r$. The physical presentation space is $\Sigma_r$, the presentation redundancy groupoid is $\Gamma_r$, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map $n_r$ is a normal-form map, not a probability law. Any physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

At least one action value must be finite, so that $Z_r > 0$. For a law on settled configurations only, restrict this construction to $N_r$, or set $S_r = +\infty$ outside $N_r$. A separately specified prior $\nu_r$ on $Q_r$ instead induces the settled law $\mu_r^{\mathrm{nf}} = (n_r)_{\#}\nu_r$. These constructions agree only if

$$\sum_{x:\, n_r(x)=q} \nu_r(x) = \mu_r(q) \qquad (q \in N_r)$$

and $\mu_r$ has no mass outside $N_r$. For example, a retraction $n(0) = 0, n(1) = n(2) = 1$ sends the uniform law on three states to weights $(1/3, 2/3)$, whereas uniform counting on its two normal forms gives $(1/2, 1/2)$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims, and a physical branch declares which one it uses.

**Normal-form projector non-selection.** For any retraction $N : Q \to Q_{\mathrm{nf}}$ onto a subset $Q_{\mathrm{nf}} \subseteq Q$, that is, any map whose restriction to $Q_{\mathrm{nf}}$ is the identity, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#}\mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on $Q_{\mathrm{nf}}$ is fixed; both statements use the retraction property. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine $X$ and its quotient-visible invariants, but they do not choose a member of $X$. If two laws $\mu, \nu$ are supported on $X$ and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite $Q$, positive $m$, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q)\exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state $\sigma_r$, self-adjoint constraints and a feasible set containing a faithful state,

$$\rho_r = \frac{\exp(\log\sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr}\exp(\log\sigma_r - \sum_a \theta_a F_{r,a})}.$$

If the feasible states lie in a proper support projection $P$, first restrict to their joint support. The same multiplier argument uses $P(\log\sigma_r)P$ and $PF_{r,a}P$, exponentiated and normalized on $P\mathcal{H}$; replacing the reference by $P\sigma_r P$ before taking its logarithm is not valid without a commutation condition. An empty feasible set has no selected state. The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse maps, with $c_{tr} = c_{sr} \circ c_{ts}$ for $t \succeq s \succeq r$. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' :\, c_{sr}(q')=q} m_s(q')e^{-S_s(q')} = \alpha_{sr} m_r(q)e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of $q$. Then

$$(c_{sr})_{\#}\mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#}\mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_{\#}\mu_n - \mu_r\|_{\mathrm{TV}} \le \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0}\left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r\right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta)\cdot F_r(q_r)$. If the residual is uniformly bounded by $\varepsilon$, the induced total-variation defect is bounded by $\tanh\varepsilon$.

**Implementation invariance and representative lifting.** If implementations $A, B$ have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_{\#}\mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_{\#}\widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel $\widetilde{P}(x, y)$ descends to $Q_r$ only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations $E_C$ are defined on $L^2(X_r, \pi_r)$, so the reference law $\pi_r$ is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant on the orthogonal complement of the common fixed space $\mathcal{F}_r = \bigcap_C \operatorname{Ran} E_C$:

$$\kappa_r = \inf_{0 \neq f \perp \mathcal{F}_r} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If every $c_C \geq \gamma_* > 0$, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0),$$

where $P_0$ projects onto $\mathcal{F}_r$. In finite dimension, $\kappa_r > 0$ whenever $\mathcal{F}_r^{\perp} \neq \{0\}$, since zero Dirichlet energy would put $f$ in every fixed range. If all states are fixed, the displayed operator inequality is vacuous. Replacing $f \perp \mathcal{F}_r$ by $f \perp 1$ requires the additional ergodicity condition $\mathcal{F}_r = \operatorname{span}\{1\}$; a conserved nonconstant record otherwise makes that infimum zero. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt. Multiplying all rates by $s > 0$ rescales the generator and gap by $s$, while leaving the fixed space, stationarity and detailed balance unchanged. Numerical recognition therefore tests identities after removing a common rate scale and restores that scale to reported gaps. An unresolved positive gap is a precision limitation, not evidence that changing clock units changed the equilibrium.

**Finite evidence accuracy.** For bounded coarse observables $O$, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to $\epsilon_{\mathrm{ref}}$, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Physical-vacuum requirement.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. A physical-vacuum interpretation requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \ge 0$, local potential $V_r$, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state $\Omega_r$, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with $\mu_r^{\mathrm{vac}}$. A continuum interpretation additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family. On the native source screens the actual density law has such a reflection-positive reconstruction along one common refining subsequence; its one-particle generator is the native continuum operator $A$ rather than $\sqrt{A}$, and it supplies no coherent preparation, relativistic vacuum or Lorentz symmetry.[3]

**Primordial and cosmological prediction firewall.** A screen covariance contains incomplete radial information. The complete one-shell map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has an infinite-dimensional kernel that persists under positivity. OPH physical primordial interpretation requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, physical mode, radial-null-space, and forward-projection receipts together with a scale-natural physical dilation intertwiner or complete radial cross-covariance tomography. A finite radial prior produces a conditional continuation. Observable CMB comparison also requires declared source, solver, dataset, covariance, nuisance, data-use, and pooled-reducer provenance. On the native source screens the continuum energy is not boundedly congruent to the static boundary

[3] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/model/observer_time_fields/REFINEMENT.md

action of a smooth compact three-dimensional Dirichlet bulk, whose Steklov eigenvalues grow like $\sqrt{n}$ while the native eigenvalues grow like $n$, and internal rank-three carrier modes vanish from bounded-energy families; a radial lift needs a radial observation algebra that the native energy does not supply.[4]

## 2 Finite Screen Spectrum Theorem Package

The screen-level spectrum theorem supplies the finite source object used by the cosmology branch. It does not supply temperature-temperature, temperature-polarization, polarization-polarization, lensing, likelihoods, or a physical cosmic microwave background interpretation. Those require the Boltzmann and data-contract theorems.

**Definition 2.1** (finite screen regulator)**.** *For regulator $r$, a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

*where $\mathcal{T}_r$ is a finite cellulation of the screen, $M_r \succ 0$ is the area/quadrature mass matrix, $\Gamma_r$ is the hidden-presentation groupoid, $\mathcal{Q}_r = \Sigma_r/\Gamma_r$ is the physical quotient, and $C_{sr}$, $J_{rs}$ are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require*

$$f^{\mathsf{T}} M_r g \to \int_{S^2} f\, g\, d\Omega$$

*with an explicit band-limited error bound*

$$\left| f^{\mathsf{T}} M_r g - \int f\, g\, d\Omega \right| \le \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

*Patch count alone is therefore not an angular-resolution certificate.*

**Definition 2.2** (geometric screen scalar)**.** *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

*and let $\bar{J}_{X,r}$ be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define*

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

*On a certified spherical chart, with $B_r = [1, n_x, n_y, n_z]$,*

$$\Pi_{<2,r} = B_r (B_r^{\mathsf{T}} M_r B_r)^{-1} B_r^{\mathsf{T}} M_r, \qquad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

*On an irregular screen, $B_r$ is replaced by the certified generalized eigenprojector onto the finite $\ell = 0, 1$ subspace. Hard-coded feature z-scores or observer-summary weights are diagnostic proxies, not $q_r$.*

**Proposition 2.3** (quotient invariance)**.** *If a recharting $U$ preserves the screen inner product and geometric readout,*

$$U^{\mathsf{T}} M_r' U = M_r, \qquad q_{0,r}' = U q_{0,r}, \qquad B_r' = U B_r,$$

*then $\Pi'_{\ge 2,r} U = U \Pi_{\ge 2,r}$, and $q_r' = U q_r$.*

[4] https://github.com/FloatingPragma/observer-patch-holography/blob/r2057/model/physical_space_time/OCCUPATION_FIELD.md

*Proof.* Substitution gives

$$\begin{aligned}\Pi'_{<2,r}U &= UB_r(B_r^{\mathsf{T}}U^{\mathsf{T}}M'_rUB_r)^{-1}B_r^{\mathsf{T}}U^{\mathsf{T}}M'_rU\\ &= UB_r(B_r^{\mathsf{T}}M_rB_r)^{-1}B_r^{\mathsf{T}}M_r = U\Pi_{<2,r}.\end{aligned}$$

Subtracting from $U$ gives the high-mode identity. □

**Definition 2.4** (physical scalar precision and action)**.** *The scalar precision $K_r$ must have a physical origin on $V_r = \operatorname{im}\Pi_{\geq 2,r}$. Allowed branches are:*

$$K_r^{\mathrm{Hess}} = \nabla^2\Phi_r\Big|_{q=0},$$

*for a quotient-visible mismatch or release free energy; a repair Dirichlet form*

$$K_r^{\mathrm{diss}} = H_r - T_r^{\mathsf{T}}H_rT_r, \qquad T_r^{\mathsf{T}}H_rT_r \preceq H_r;$$

*or the reversible generator form*

$$K_r^{\mathrm{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger_H}).$$

*A raw nonsymmetric repair matrix or caller-supplied $\kappa$ is not a precision operator. The absolute normalization must be fixed independently, for example by*

$$K_0Y_{\ell m} = \ell(\ell+1)Y_{\ell m}$$

*on the $\theta = 0$ branch. Once normalized,*

$$S_{\mathrm{scr},r}[q] = \frac{1}{2A_{q,r}}\langle q, K_rq\rangle_r = \frac{1}{2A_{q,r}}q^{\mathsf{T}}M_rK_rq.$$

**Theorem 2.5** (finite MaxEnt screen covariance)**.** *Let $K_re_i = \kappa_ie_i$ with an $M_r$-orthonormal basis of $V_r$, $\kappa_i > 0$, and $d_r = \dim V_r$. Among continuous densities in coefficients $q = \sum_i a_ie_i$ with $\mathbb{E}[a_i] = 0$ and*

$$\frac{1}{2}\mathbb{E}\langle q, K_rq\rangle_r = E_{q,r}^{\mathrm{src}},$$

*the unique entropy maximizer is*

$$d\mu_r(q) = Z_r^{-1}\exp\left[-\frac{1}{2A_{q,r}}\langle q, K_rq\rangle_r\right]dq, \qquad A_{q,r} = \frac{2E_{q,r}^{\mathrm{src}}}{d_r},$$

*and*

$$\mathbb{E}[a_ia_j] = \delta_{ij}\frac{A_{q,r}}{\kappa_i}.$$

*Proof.* The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature $\beta_r$. The expected energy is

$$E_{q,r}^{\mathrm{src}} = \frac{1}{2}\sum_i \kappa_i\frac{1}{\beta_r\kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus $A_{q,r} = \beta_r^{-1} = 2E_{q,r}^{\mathrm{src}}/d_r$. Strict concavity of entropy gives uniqueness. □

*Remark* 2.6 (discrete quotient branch). For a discrete finite quotient with intrinsic base weight $m_r(q)$, the exact Gibbs law is

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

**Definition 2.7** (source release energy). *Let $m_r^0$ be the tracially pointed base weight on the finite physical collar quotient. Let $F_{r,a}$ be a finite ledger of primitive collar observables: scalar occupancy, protected-center presence, released volume, conserved source charges, and the release clock. The ledger excludes screen energy, sky spectra, CMB residuals, likelihoods, fitted amplitudes, and measurement-calibrated proxies. For source-emitted constraint values $c_{r,a}^{\mathrm{col}}$, the selected release law is*

$$\nu_r^{\mathrm{rel}}(x) = \frac{m_r^0(x) \exp[-\sum_a \lambda_{r,a} F_{r,a}(x)]}{Z_r}, \qquad \mathbb{E}_{\nu_r^{\mathrm{rel}}} F_{r,a} = c_{r,a}^{\mathrm{col}}.$$

*Strict concavity of relative entropy gives uniqueness on full support. A boundary optimum uses the same form on its certified support. The constraint ledger, base weight, source DAG, multipliers, support, and refinement maps belong to the receipt.*

*For this independently selected quotient ensemble on released collar normal forms, define*

$$E_{q,r}^{\mathrm{src}} = \frac{1}{2} \int \langle q_r(x), K_r q_r(x) \rangle_r \, d\nu_r^{\mathrm{rel}}(x).$$

*The receipt must expose $\nu_r^{\mathrm{rel}}$, the base measure, $K_r$, $d_r$, $E_{q,r}^{\mathrm{src}}$, $A_{q,r}$, no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.*

**Proposition 2.8** (microscopic-law necessity). *For fixed positive $K_r$, the Gaussian family $Z(A)^{-1} \exp[-\langle q, K_r q \rangle/(2A)]$ exists for every $A > 0$. The quadratic MaxEnt form therefore leaves one positive scale free. The source release law in Definition 2.7, or an equivalent source-only vacuum law, supplies the energy that fixes $A_{q,r} = 2E_{q,r}^{\mathrm{src}}/d_r$.*

**Theorem 2.9** (rotational screen spectrum). *If the continuum precision $K_\theta$ commutes with the $SO(3)$ action on scalar functions, then by Schur's lemma each $\mathcal{H}_\ell$ is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \qquad \ell \geq 2.$$

*Then $\Lambda_\ell(0) = \ell(\ell+1)$, and the MaxEnt covariance gives*

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

*Consequently $\mathcal{D}_\ell^q = \ell(\ell+1) C_\ell^q/(2\pi) \sim (A_q/2\pi) \ell^{-\theta}$.*

**Definition 2.10** (source-derived conformal precision). *Let $L_r \succeq 0$ be the normalized detailed-balanced scalar collar Dirichlet operator on $V_r$, with continuum limit $-\Delta_{S^2}$. Set*

$$B_r = (L_r + \tfrac{1}{4} I)^{1/2}, \qquad K_{\theta,r} = \frac{\Gamma(B_r + \frac{3}{2} + \frac{\theta}{2})}{\Gamma(B_r - \frac{1}{2} - \frac{\theta}{2})}.$$

*For $-2 < \theta < 4$ on the retained $\ell \geq 2$ branch, this operator is positive. The gamma recurrence gives $K_{0,r} = L_r$ exactly. The finite receipt carries detailed balance, positivity, the $\ell(\ell+1)$ normalization, anisotropy splitting, and refinement residuals.*

**Theorem 2.11** (finite refinement error)**.** *Suppose that for* $2 \le \ell \le L$

$$\left|\frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1\right| \le \varepsilon_{K,r}(L), \qquad \left|\frac{A_{q,r}}{A_q} - 1\right| \le \varepsilon_{A,r}, \qquad \varepsilon_{K,r} < 1.$$

*Then* $C^q_{\ell m,r} = A_{q,r}/\lambda_{\ell m,r}$ *satisfies*

$$\left|\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell(\theta)} - 1\right| \le \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

*Proof.* Write $A_{q,r} = A_q(1 + a_r)$ and $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$. Then

$$\frac{C^q_{\ell m,r}}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from $|a_r| \le \varepsilon_A$, $|b_{\ell m,r}| \le \varepsilon_K$. □

**Theorem 2.12** (refinement-semigroup tilt)**.** *Let* $R_b$ *be the scalar refinement map for scale ratio* $b > 1$*. If one isolated scalar covariance mode has positive eigenvalue* $\lambda(b)$*,* $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$*, and* $\lambda$ *is continuous, then there is a unique real* $\theta$ *with*

$$\lambda(b) = b^{-\theta}, \qquad \theta = -\frac{\log \lambda(b)}{\log b}.$$

*Proof.* Set $g(t) = -\log \lambda(e^t)$. The semigroup law gives $g(t + s) = g(t) + g(s)$. Continuity makes $g(t) = \theta t$, hence the result. □

**Definition 2.13** (edge-center reserve generator receipt)**.** *Write* $s = \log b$ *for logarithmic refinement thickness. Let* $u_{\mathrm{full}}(s)$ *be the scalar-conditioned covariance-survival cocycle across a full oriented collar, and let* $u_q(s)$ *be its source-facing half-collar restriction. The receipt requires*

$$u(s + t) = u(s)u(t), \qquad u(0) = 1,$$

*strong continuity at zero, the source-derived full-collar density*

$$-u'_{\mathrm{full}}(0) = \frac{P_\star}{24},$$

*and the orientation-reversal coarea identity*

$$-u'_q(0) = \frac{1}{2}[-u'_{\mathrm{full}}(0)] = \frac{P_\star}{48}.$$

*These are infinitesimal generator statements. A one-step survival probability has exponent* $-\log u_q(\log b)/\log b$*.*

**Theorem 2.14** (edge-center tilt and repair-clock reconciliation)**.** *Under Definition 2.13,*

$$u_q(s) = e^{-\theta s}, \qquad \theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}.$$

*In the coordinate* $\theta = \kappa_{\mathrm{rep}}(P_\star - \varphi)$*, the same branch has*

$$\kappa^{\mathrm{edge}}_{\mathrm{rep}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

*The value* $e$ *is a separate diagnostic hypothesis unless an additional identity equates it with this source-derived coordinate.*

*Proof.* For $g(s) = -\log u_q(s)$, the cocycle law gives the continuous Cauchy equation. Hence $g(s) = \theta s$. Differentiation at zero and the half-collar identity give $\theta = P_\star/48$. The formula for $\kappa_{\mathrm{rep}}^{\mathrm{edge}}$ is algebraic. □

**Theorem 2.15** (homogeneous radial source family)**.** *Assume source-stress closure, a single clock, freezeout, multicenter consistency, and translation/rotation covariance. Let $C_\zeta = M_{\Delta_\zeta^2}$ on the common physical $d\ln k$ mode basis. If a scale-natural source embedding transports refinement to physical dilation and its finite operator residual converges to*

$$D_s^{-1} C_\zeta D_s = e^{-\theta s} C_\zeta, \qquad (D_s f)(k, \hat{k}) = f(e^{-s}k, \hat{k}),$$

*then every positive measurable representative satisfies*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}$$

*almost everywhere. Continuity gives the pointwise identity. A single shell has an infinite-dimensional radial kernel, including positive ambiguities; complete radial cross-covariances give the independent spherical-Hankel tomography route.*

*Proof.* Conjugation of the multiplication operator gives $\Delta_\zeta^2(e^s k) = e^{-\theta s}\Delta_\zeta^2(k)$ almost everywhere. In logarithmic coordinates, rational-translation invariance on a common full-measure set forces $e^{\theta t}\Delta_\zeta^2(e^t)$ to be constant almost everywhere. The one-shell and tomography statements follow from the correlation-restriction and spherical-Hankel theorems in the primordial bridge packet. □

**Theorem 2.16** (thin-shell power-law lift)**.** *Assume $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$ and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

*Then*

$$C_\ell^q = 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) j_\ell^2(kR_\star),$$

*and, for $-2 < \theta < 4$,*

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1+\theta/2)}{\Gamma(3/2+\theta/2)}.$$

*Thus*

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}.$$

**Proposition 2.17** (finite-window bound)**.** *Let $\Psi_\ell(k) = \int dr\, W(r) j_\ell(kr)$, $W \geq 0$, $\int W dr = 1$, with mean $R_\star$ and variance $\sigma_R^2$. If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \operatorname{supp} W} |j_\ell''(kr)|,$$

*then*

$$|C_{\ell,W}^q - C_{\ell,\mathrm{shell}}^q| \leq 4\pi Z_\star^2 \int d\ln k\, \Delta_\zeta^2(k) \left[2\delta_\ell(k) + \delta_\ell(k)^2\right].$$

**Proposition 2.18** (radial non-identifiability)**.** *For a finite radial basis, $C = Ap$. If $A \in \mathbb{R}^{N_\ell \times N_k}$ has rank $r$, then $\dim\ker A = N_k - r$, and every $p + v$ with $v \in \ker A$ gives the same screen spectrum. A physical radial reconstruction therefore requires either a source theorem restricting $p$, or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

**Theorem 2.19** (promotion contract for the thin-shell power-law pipeline)**.** *Under the declared thin-shell lift and power-law source family above, one hash-locked source DAG may emit*

$$(q_r, S_{\mathrm{scr},r}, \theta, A_{q,r}, C^q_{\ell,r}, A_{\zeta,r}, \Delta^2_{\zeta,r})$$

*as a conditional primordial packet when the geometric-scalar, collar-law, release-energy, reserve-generator, conformal-precision, refinement, source-stress, single-clock, freezeout, adiabaticity, isocurvature, phase-coherence, thin-shell, radial-null, finite-window, and forward- residual receipts pass on that DAG. Any measurement, likelihood, fitted parameter, or measurement-calibrated ancestor fails the source packet. Physical temperature and polarization spectra require the independent Boltzmann, recombination, nuisance, covariance, and likelihood gates. This contract is specific to the displayed pipeline. A different source-only radial reconstruction or forward projection may use another fully typed contract, provided it certifies its own injectivity or null space, window and convergence control, forward residual, and no-target-leak provenance.*

**Target 2.20** (requirements for a source-derived screen spectrum)**.** *A concrete source-derived spectrum requires the following receipt families before primordial interpretation:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *the infinitesimal reserve generator, scalar refinement tilt, and angular screen spectrum with a finite error budget;*
4. *source-stress, clock, freezeout, radial-window, radial-null, forward-residual, and transfer firewall receipts.*

*The theorem and algorithm packet is complete for the declared thin-shell power-law pipeline. No finite source DAG that passes this receipt set is supplied, and the list is not a no-go for a differently typed radial pipeline.*

**Native source and the screen premises.** The conformal precision of Definition 2.10 takes the round continuum limit $-\Delta_{S^2}$ as a premise. The native repair law supplies a continuum generator $A$ on the screen whose energy lies between fixed multiples of the gradient energy of the flat comparison polyhedron; its identification with $-\Delta_{S^2}$ is not established [6]. The executable source also supplies no source-only vacuum law in the sense of Proposition 2.8: its full configuration reference approaches the normalized trace, and its continuous wave readout has the classical displacement covariance $A^{-1}$, not the vacuum covariance $(2\sqrt{A})^{-1}$ [7]. Under any bounded form-congruent identification of the shell precision $K_\theta$ with the native long-average field, the min-max argument of the boundary-trace test [6] matches the eigenvalue growth $n^{1+\theta/2}$ of $K_\theta$ against the linear growth of $\lambda_n(A)$ and forces $\theta = 0$, so the native field's spatial covariance cannot carry the edge-center tilt, which requires a separate ingredient.

## 3 Branch Thesis

The branch requirement is:

| An inflation-free finite-source branch would have to replace the logical jobs of inflation by clock-slice spatial-holonomy selection, same-boundary or low-$k$ repair coherence, a geometric screen spectrum, MaxEnt release, and ordinary Boltzmann transfer. |
|---|

This paper states those jobs as premises. Apart from the record-density expansion law of Section 6, it supplies none of them: flatness is identified but not selected, and horizon coherence has no derivation (Section 9). The claim becomes predictive only when the source objects are fixed before observational comparison. The proposed screen synchronization must be related to a declared state–algebra pair. Its modular group supplies a dimensionless parameter; the local clock readout and the cosmological-duration identification require their own premises. In the executable observer, passive native repair cannot act as a nontrivial modular automorphism, an active event clock executes one compiled modular step per incident native event with declared coherent services [5], and equal mean rates of local event clocks give a stochastic relative-clock law. No universal synchronization of the screen follows from these constructions.

The dark continuation of this branch is unspecified until a finite source fixes the anomaly abundance $\bar{\rho}_A(a)$ and a finite covariant dark-stress parent supplies $w_A$, $c^2_{s,A}$, $\sigma_A$, $Q^\mu_A$, recipient stress and the cold-dark-matter limit; the local scalar-channel identity of coherent-matter same-channel forcing does not supply that parent.

# 4 Physical Conditions

| Physical question | Required result or premise |
|---|---|
| Finite early state | Quotient normal-form branch with declared base measure and implementation invariance. |
| Flatness | Zero clock-slice spatial Levi–Civita holonomy identifies $\kappa = 0$. Exact selection requires a derivation from the finite screen or an explicit curvature premise. Damping gives only an approximate $\lvert K\rvert$ or $\lvert\Omega_K\rvert$ bound. |
| Horizon coherence | Same-boundary unique-extension theorem or a uniform low-$k$ repair gap. The native screen generator has lowest eigenvalue at least $3/4096$ only in the rescaled attempt parameter $\tau = \text{attempts}/(1260 \cdot 16^l)$, so per native attempt its slowest relaxation rate falls as $16^{-l}$; it is a two-dimensional spatial spectrum, not a physical low-$k$ gap [6, 7]. |
| Near scale invariance | Geometric screen scalar, normalized scalar precision, source release energy, infinitesimal reserve-generator tilt, and angular screen-spectrum receipts. |
| Amplitude | The conditional theorem fixes $A_q = 2E^{\text{src}}_q/d$ and its source-family conversion to $A_\zeta$. A finite collar release law and energy receipt must supply their values without CMB data. |
| Hot start | MaxEnt release into the realized radiation branch with a common release clock. |
| Expansion history | Conformal invariance of the layered order and the $a^4$ record-density weight of Section 6; the record-production profile is a supplied datum, with the fixed-capacity de Sitter profile $a = -1/(H\eta)$ as one inhabitant. |
| Acoustic transfer | Declared source functions and a Boltzmann transfer calculation. |

# 5 Numerical Acceptance Conditions

The simulator source and its public data archives are cited in [4]. For this paper the simulator may emit diagnostic screen spectra, observer-screen synchronization receipts, and release-amplitude candidates. It supports an inflation-replacement interpretation only when one finite source supplies the primordial growth mode, physical radial lift, transfer calculation, and likelihood inputs. The executable observer construction emits none of these [5].

# 6 Expansion as Record Density

The spacetime construction supplies a flat causal-order and count-volume limit for the conservative golden-record population and the retrospective count clock. The following statements transfer that family to a spatially flat expanding background. They are exact and machine-checked in the Lean module named below; the profile law that selects how many records a comoving cell produces per conformal tick is a supplied datum, and no statement below derives the Friedmann equation. Conformal time $\eta$, the layer index $j$ and the durations $\sigma_j\Delta$ called physical ticks are coordinates of the supplied FLRW chart; none is a calibrated observer clock.

## 6.1 Expansion as record density on the source order

The spatially flat Friedmann–Lemaître–Robertson–Walker metric in conformal time $\eta$ is

$$ds^2 = \sigma(\eta)^2(c^2\,d\eta^2 - |dx|^2), \tag{1}$$

with scale factor $\sigma(\eta) > 0$; the letter $a$ is the edge radius of the read law. This metric is a positive conformal factor times the flat metric in the coordinates $(\eta, x)$. A positive conformal factor leaves the causal order of a Lorentzian metric unchanged, and the causal order fixes the metric up to such a factor (Hawking, King and McCarthy 1976; Malament 1977). On the spatially flat family the conformal factor is the entire background, so every profile $\sigma$ has the causal order of flat space in comoving coordinates. The layered read law of the source-net construction places an event at every $(j, s)$ for a site $s$ in a population $S$, lets $(j+1, t)$ read $(j, s)$ whenever $\|t - s\| \le a$, and takes the reflexive transitive closure of the reads as the order. Written in comoving coordinates with the tick $\Delta = a/c$ in conformal time, this law generates one finite order for every profile. The profile enters only through the count measure.

**Proposition 6.1** (Conformal invariance of the layered order)**.** *Let $S$ be a subset of a real normed space, $a > 0$ an edge radius and $\sigma_j > 0$ the scale at layer $j$. The physical position of the site $s$ at layer $j$ is $\sigma_j s$ and the physical read radius at that layer is $\sigma_j a$. The physical read from $(j, s)$ to $(j+1, t)$,*

$$\|\sigma_{j+1}t - \sigma_{j+1}s\| \le \sigma_{j+1}a,$$

*holds exactly when $\|t - s\| \le a$. The layered orders generated by the physical reads and by the comoving reads coincide, for every profile $(\sigma_j)_{j\ge 0}$.*

*Proof.* For $\sigma > 0$, $\|\sigma t - \sigma s\| = \sigma\|t - s\|$, and multiplication by $\sigma$ preserves and reflects the comparison with $\sigma a$. The two read relations are equal, hence so are their reflexive transitive closures. □

**Proposition 6.2** (FLRW count measure)**.** *Assign a comoving cell of volume $v_s$ to every site $s$ and the conformal tick $\Delta$ to every layer. Under a constant physical event density $\rho$, the physical tick at*

*layer $j$ is $\sigma_j\Delta$ and the physical cell volume is $\sigma_j^3 v_s$, so a finite set $I$ of events carries the mass*

$$M_\sigma(I) = \rho \sum_{(j,s)\in I} (\sigma_j\Delta)(\sigma_j^3 v_s) = \rho \sum_{(j,s)\in I} \sigma_j^4\, \Delta\, v_s, \tag{2}$$

*the discrete form of the four-volume weight $\sigma^4\, d\eta\, d^3x$. Write $M_1(I) = \rho\Delta \sum_{(j,s)\in I} v_s$ for the flat mass. If $0 \le \sigma_{\min} \le \sigma_j \le \sigma_{\max}$ on the layers of $I$, then*

$$\sigma_{\min}^4 M_1(I) \le M_\sigma(I) \le \sigma_{\max}^4 M_1(I). \tag{3}$$

*A constant profile $\sigma_j = \sigma$ gives $M_\sigma(I) = \sigma^4 M_1(I)$.*

*Proof.* The identity in (2) is the product of the two scaled factors. Fourth powers are monotone on nonnegative reals and $\Delta v_s \ge 0$, so each term of the sum lies between the same term with $\sigma_j$ replaced by $\sigma_{\min}$ and by $\sigma_{\max}$. Summing proves (3); the constant case is the common factor $\sigma^4$. □

**Corollary 6.3** (Count clock in an expanding family)**.** *Let $I$ and $J$ be finite event sets with uniform cells, $J$ nonempty, with $\sigma_{\min}^I \le \sigma_j \le \sigma_{\max}^I$ on the layers of $I$, $\sigma_{\min}^J \le \sigma_j \le \sigma_{\max}^J$ on the layers of $J$, $\sigma_{\min}^I \ge 0$ and $\sigma_{\min}^J > 0$. Then*

$$\frac{\sigma_{\min}^I}{\sigma_{\max}^J} \left(\frac{|I|}{|J|}\right)^{1/4} \le \left(\frac{M_\sigma(I)}{M_\sigma(J)}\right)^{1/4} \le \frac{\sigma_{\max}^I}{\sigma_{\min}^J} \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{4}$$

*Let $I$ and $J$ approximate comoving diamonds of conformal durations $T_I$ and $T_J$ in the sense of the count-clock theorem for the flat order, so that $(|I|/|J|)^{1/4} \to T_I/T_J$ under refinement. A comoving worldline through a diamond of conformal duration $T$ accrues proper time between $\sigma_{\min} T$ and $\sigma_{\max} T$. In the refinement limit the clock read from the physical masses, $(M_\sigma(I)/M_\sigma(J))^{1/4}$, and the proper-time ratio $\tau_I/\tau_J$ lie in one interval whose endpoints have the quotient $(\sigma_{\max}^I/\sigma_{\min}^I)(\sigma_{\max}^J/\sigma_{\min}^J)$; the two readings agree up to this factor in either direction. For a profile continuous in $\eta$ and diamonds shrinking to points, the factor tends to one and the clock reads the proper-time ratio exactly in the limit.*

*Proof.* Divide the two sandwich inequalities (3) for $I$ and $J$, with $M_1(I)/M_1(J) = |I|/|J|$ for uniform cells, and take the increasing fourth root; the fourth root of $\sigma^4$ is $\sigma$. The proper time along the comoving worldline is $\int \sigma\, d\eta$ over the conformal duration, bounded by the extreme scales times $T$; the ratio of two such enclosures has the stated endpoint quotient. Continuity of $\sigma$ forces $\sigma_{\max}/\sigma_{\min} \to 1$ on shrinking diamonds. □

**Proposition 6.4** (Finite expanding diamonds need not read proper time)**.** *For the supplied radiation profile $\sigma(\eta) = 1 + \eta$ and comoving vertical diamonds starting at $\eta = 0$, take conformal durations $1$ and $1/2$. Their continuum mass ratio is*

$$\frac{M_\sigma(1)}{M_\sigma(1/2)} = \frac{1516288}{44451} \ne \left(\frac{12}{5}\right)^4 = \left(\frac{\tau(1)}{\tau(1/2)}\right)^4.$$

*Thus their fourth-root volume clock differs from the proper-time ratio even after arbitrarily fine discretization at these fixed durations.*

*Proof.* The common factor $4\pi\rho c^3/3$ cancels from the two integrals

$$M_\sigma(T) \;\propto\; \int_0^{T/2} (1+\eta)^4 \eta^3\, d\eta + \int_{T/2}^{T} (1+\eta)^4 (T-\eta)^3\, d\eta.$$

Expanding the polynomials and integrating gives the displayed rational ratio. Along the comoving worldline, $\tau(T) = \int_0^T (1+\eta)\, d\eta = T + T^2/2$, hence $\tau(1)/\tau(1/2) = 12/5$. Subtracting fourth powers gives the strictly positive rational $25944064/27781875$. □

The Lean module `SourceNetClockSeparation` checks the evaluated mass polynomial and rational mismatch; the identification of that polynomial with the geometric integral is the analytical argument above. The resulting relative clock bias is $+0.696366\%$. It is a continuum finite-window effect, not a discretization error. For the supplied profiles $\sigma(\eta) = (1-\eta/2)^{-1}$ and $\sigma(\eta) = [1+(\sqrt{2}-1)\eta]^2$, independent split quadratures give $-1.595321\%$ and $+0.265422\%$ at the same durations. The shrinking-diamond hypothesis in Corollary 6.3 is therefore substantive. The exact polynomial control and numerical receipts accompany the observer-dynamics verifier. No expansion profile is selected by this calculation.

**Corollary 6.5** (Redshift as a count ratio)**.** *Let two congruent comoving diamonds, with the same comoving cells and the same conformal ticks, lie at the emission epoch, of scale $\sigma_e$, and at the reception epoch, of scale $\sigma_0$, the profile being constant across each. Their masses are $n_e = \kappa\sigma_e^4$ and $n_0 = \kappa\sigma_0^4$ with the common factor $\kappa = \rho\Delta\sum_{(j,s)\in I} v_s$, and*

$$1 + z = \frac{\sigma_0}{\sigma_e} = \left(\frac{n_0}{n_e}\right)^{1/4}. \tag{5}$$

*Proof.* The constant case of Proposition 6.2 gives both masses; $\kappa$ cancels in the ratio, and the positive fourth root of $(\sigma_0/\sigma_e)^4$ is $\sigma_0/\sigma_e$. □

*Remark* 6.6 (Expansion history as a record-production law). On this family the scale-factor profile is a statement about record production: a comoving cell of volume $v$ produces $\rho\,\sigma(\eta)^4\,\Delta\,v$ records per conformal tick, and a profile $\sigma(\eta)$ is the same datum as a record-production rate per comoving cell. The de Sitter profile $\sigma(\eta) = -1/(H\eta)$ on $\eta < 0$ solves $d\sigma/d\eta = H\sigma^2$, and its record density per comoving cell is $n(\eta) \propto (H\eta)^{-4}$; the fixed-capacity branch of the corpus, $w = -1$, corresponds to this profile. The Lean module `SourceNetConformalRecordDensity` checks the order invariance, the mass identity and sandwich, the clock enclosure (4), the redshift identity and the de Sitter derivative and density identities over the reals.

Three items lie outside this construction. The source law selecting the profile, the record-production rate per comoving cell, is not derived; the profile is a supplied datum. A reweighted flat order determines at most the conformal factor; on the spatially flat family that factor is the entire background, and spatial curvature $k \neq 0$ changes the comoving spatial metric, which this construction does not treat. No physical clock and no population selection by native repair follow from these identities.

# 7 Flatness Boundary

The flatness branch is split into identification and selection. On a clocked Friedmann–Lemaître–Robertson–Walker (FLRW) continuation with $u_a = -N\nabla_a\chi$, $h_{ab} = g_{ab} + u_a u_b$, and $K(\tau) = \kappa/a(\tau)^2$, vanishing area-normalized small-loop holonomy of the spatial Levi–Civita connection identifies $K = 0$ and hence $\kappa = 0$. It does not select that sector from the raw FLRW fiber. A numerical calculation must distinguish exact selection derived from the finite screen, a flat branch conditioned on an explicit premise, and a branch for which no selection theorem is supplied. Screen and collar defect decay is a repair diagnostic rather than a spatial curvature holonomy. Curvature damping supplies only an approximate bound on $|K|$ or $|\Omega_K|$.

## 8 Screen-Spectrum Branch

The load-bearing screen result is the angular screen covariance. It is distinct from a CMB spectrum:

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

This statement requires the geometric $q_r$ of Definition 2.2, a normalized conformal precision operator $K_{\theta,r}$, the primitive collar release law, $E_{q,r}^{\mathrm{src}}$, the MaxEnt covariance theorem, and a finite-to-continuum refinement error budget. The edge-center theorem gives

$$\theta = \frac{P_\star}{48}, \qquad n_s = 1 - \frac{P_\star}{48}, \qquad \kappa_{\mathrm{rep}}^{\mathrm{edge}} = \frac{P_\star}{48(P_\star - \varphi)}.$$

Its premise is a source-emitted full-collar generator density $P_\star/24$ with the orientation-half identity. A finite one-step survival probability is converted through $-\log u/\log b$. The $e(P_\star - \varphi)$ clock branch is a diagnostic alternative. The finite generator and clock receipt is not supplied here. No finite source emits the density $P_\star/24$; it is a declared branch input. The tilt therefore has the same status as the capacity display of the cosmological constant: a declared one-parameter map evaluated at a pixel that the measured fine-structure constant locates. Its numerical content is the declared density, since the cocycle form alone admits every exponent (see the end of this section).

At the comparison pixel $P_\star = 1.6309682094$, located from the measured fine-structure constant, the edge-center value is $n_s = 0.96602$; the forward pixel 1.6309721 moves it by less than $10^{-7}$. The retrospective comparison with public temperature and polarization data is recorded in the cosmology postdiction ledger released with these papers: the value lies $0.27\sigma$ above Planck 2018, $0.69\sigma$ above SPT-3G with Planck, $0.79\sigma$ below the SPT, Planck and ACT combination, and $1.28\sigma$ below ACT DR6 with Planck. The diagnostic $e(P_\star - \varphi)$ branch gives 0.96484. The official joint posterior chains of ACT DR6 with Planck, lensing and DESI 2024 baryon acoustic oscillations, whose mean is $0.9743 \pm 0.0034$, place the edge-center branch at their lower 0.612 percent position and the diagnostic branch at 0.200 percent; the measured tail differs from the Gaussian summary [10, 9]. Those data precede the registration, so this is a retrospective tension under baseline ΛCDM, not a likelihood ratio, and the registered comparison with a later dataset is unchanged. These are seen-data comparisons of a receipt-conditional value and select neither branch. The operative tilt protocol is FZ-15, frozen on 2026-10-04 with register status `frozen_stamped_upgrade_pending`; its complete combination, pivot, thresholds and anchoring requirements govern the prospective comparison. On the same branch the dilation cocycle gives zero running, the rank-one single-clock normal form gives zero source-side isocurvature. The scalar screen field emits no transverse-traceless source, but that condition alone leaves homogeneous tensor modes free. A zero linear primordial tensor amplitude additionally requires zero initial tensor amplitude and velocity and no tensor forcing; those initial data are extra branch premises. Planck 2018 gives a running of $-0.0045 \pm 0.0067$ and ACT DR6 with Planck, lensing and baryon acoustic oscillations $0.0062 \pm 0.0052$; BICEP/Keck 2018 bounds $r_{0.05} < 0.036$ and the fourth Planck release with BK18 and BAO $r < 0.032$; Planck 2018 bounds the uncorrelated cold-dark-matter isocurvature fraction at 3.5 to 3.9 percent. A tensor null at a stated sensitivity excludes only specified models predicting a larger signal, since slow-roll inflation allows arbitrarily small tensor amplitudes. It cannot establish exact zero or distinguish this branch from all single-field models. The running-zero row likewise tests only alternatives resolved by its measurement errors. The exponential form of the survival cocycle and the tilt $P_\star/48$ from the half-collar density, together with the exact rational brackets of every comparison above, are machine-checked in the Lean modules `EdgeCenterTiltCocycle` and `CosmologyLedgerBrackets`.

Under the declared source dilation cocycle, the primordial power law

$$\Delta_\zeta^2(k) = A_\zeta \left(\frac{k}{k_\star}\right)^{-\theta}$$

is the unique member of the one-dimensional source family. The exact thin-shell identity fixes the amplitude conversion

$$A_\zeta = \frac{A_q}{\pi^{3/2}} \frac{\Gamma(3/2+\theta/2)}{\Gamma(1+\theta/2)}$$

at the source pivot when $q = \Pi_{\geq 2}\zeta(R_\star \hat{n})$ and $k_\star R_\star = 1$. For the general attachment $q = Z_\star \Pi_{\geq 2}\zeta(R_\star \hat{n})$, the displayed expression is divided by $Z_\star^2 (k_\star R_\star)^\theta$, with $Z_\star \neq 0$ and $-2 < \theta < 4$. A finite run requires $R_\star$, the radial window, the unrestricted null-space report, the forward residual, and the physical scale. The screen theorem proves no temperature, polarization, lensing, acoustic-peak, or likelihood statement.

### 8.1 What the native observer records determine

The fair balanced integer repair law supplies an absolute spectrum for a specified load readout, without fitting an observed sky. On a connected weighted port graph with $p$ ports and $r$ raised loads, stationary continued repair, in the continuous native attempt parameter $t$ at unit attempt rate per seam, gives

$$\mathrm{Cov}(z(t), z(0)) = \kappa e^{-|t|L/2}\Pi, \qquad \kappa = \frac{r(p-r)}{p(p-1)}, \quad \Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p.$$

For the ordinary local average $f_T = T^{-1}\int_0^T z(t)\,dt$,

$$\mathrm{Var}(v_\lambda^{\mathsf{T}} f_T) = \kappa \frac{2(u-1+e^{-u})}{u^2}, \qquad u = \lambda T/2.$$

The fixed-graph fluctuation record $T^{-1/2}\int_0^T z(t)\,dt$ instead converges to a Gaussian law with covariance $4\kappa L^+$. Incident pre-attempt records, including null attempts and centered with the prepared filling, have the same leading inverse-Laplacian term with operator correction at most $3\kappa/d_{\min}$, at a common native attempt parameter; $t$ is not a calibrated observer clock. These are analytical consequences of actual local port interaction and readback, checked by finite-state enumeration and independent moment evolution. The detailed proofs and the distinct occupation and event normalizations are given in the screen-microphysics companion [2].

The amplitude is in native load-record units. Occupancy, attempt clock, duration, gain and stationary preparation are specified inputs. For a centered geometric readout $q = Bz + g$, orthogonal projection in the stationary law gives at every finite duration

$$C_T(q) = BC_T(z)B^{\mathsf{T}} + C_T(g), \qquad C_T(g) \succeq 0.$$

Thus a measured collar-volume readout can test its density overlap; the load record cannot be substituted for that geometric observable. Along one common subsequence of refining source screens with $240 \cdot 4^l$ ports, one-third filling from the declared uniform fixed-number preparation, independent uniform seam selection and fair repair, normalized native occupation snapshots read at native attempt count $\lfloor 1260 \cdot 16^l \tau \rfloor$ converge jointly to a stationary Gaussian family with covariance $\langle f, e^{-|\tau-u|A} g\rangle$. Here $A$ is one positive self-adjoint generator whose form is a conservative, regular, strongly local Dirichlet form of heat-trace spectral dimension two [6, 7]. These finite-dimensional

laws assert no path-space tightness or spatial convergence rate, and the density field lives on the two-dimensional screen readout; it is not identified with a primordial curvature field. With fixed-area equal port weights and a common deterministic filter of bounded $L^1$ gain $W$, the ensemble mean empirical harmonic pseudo-power satisfies $\widetilde{C}_\ell = (2\ell+1)^{-1}\sum_m \mathbb{E}|a_{\ell m}|^2 \le \pi W^2/(p-1)$, even without exact quadrature. This is not a pointwise bound on every sky realization. Ordinary averages have $W = 1$; amplified fluctuation records do not have uniformly bounded gain as their duration grows.

There are two additional quantitative restrictions. A common deterministic temporal filter of stationary density obeys $d\log V/d\log\lambda \ge -1$, so under a separately established three-dimensional $\lambda \propto k^2$ identification it cannot produce the required $k^{-3}$ curvature spectrum. Under the same operator dictionary that identification fails for the native continuum generator: its eigenvalues obey $n/49920 \le \lambda_n(A) \le 3n/2$, which excludes a unitary identification, up to a fixed scale and bounded mass shift, with a second-order elliptic operator on a smooth compact three-dimensional manifold, and excludes a boundedly invertible energy congruence with the static boundary action of a smooth compact three-dimensional Dirichlet bulk [6]. The native port operator is also not proved round: at level four the seven-mode band splitting is 14.642%, compared with 0.01561% for a supplied round-metric operator. An inverse graph covariance retains its precision's anisotropy.

Absolute inherited/fresh/settled covariance is recovered for all 61 retained refinement chains and eight group sizes. For unshuffled level-eight to level-nine chains at 65,536 carriers per group, the inherited-only residual variance is $2.55666955 \times 10^{-6}$; removing the linear fresh-load response leaves $4.00020779 \times 10^{-8}$. The first variance times group size is 0.16755, close to the supplied fresh variance 1/6. Nested same-location residuals satisfy $\mathrm{Cov}(e_M, e_N) = \mathrm{Var}(e_{\max(M,N)})$; this is a projection identity, not a separated-position covariance or independence theorem. Group-dependent inherited responses do not supply a common refinement generator. The retained sample shares initial fields, and fresh reconstruction depends on pinned producer and random-number semantics; it supplies neither $A_s$ nor $n_s$. The specifications, original chain receipts, independent controls and absolute-covariance extraction are archived [12].

**Refinement survival does not select a numerical exponent.** For every real $\theta$, $u_\theta(s) = e^{-\theta s}$ is a continuous multiplicative survival cocycle. For $\theta \ge 0$ it also obeys $0 < u_\theta(s) \le 1$ on $s \ge 0$. Distinct exponents therefore inhabit the same cocycle interface; its type alone cannot select $P_\star/48$. The Lean module `EinsteinBranch.EdgeCenterTiltFreedom` [11] checks this freedom against the canonical cocycle definition. Numerical tilt selection requires the source generator and its physical dilation attachment. Neither nested regression orthogonality nor coarse load survival supplies that derivative premise.

## 9 Unsupplied Premises

The same-boundary or low-$k$ coherence branch has no quotient-language derivation. A finite source emitting the geometric collar-volume field

$$q_r = \Pi_{\ge 2,r}(1/3)\log(J_{X,r}/\bar{J}_{X,r})$$

is not constructed by the stated source data. The same source must supply the primitive collar release law and pooled release energy that fix $A_q$, together with the full-collar infinitesimal reserve generator and orientation-half condition used by the conditional $P_\star/48$ theorem. Its source-dependency graph must contain compatible stress, clock, freezeout, physical-scale, radial-window,

null-space, and forward-residual evidence; this scientific prerequisite graph is distinct from the semantic read-from order of the consensus layer [3]. The resulting quantities are the inputs to the cosmic microwave background transfer and likelihood calculations.

The rank-three source Gram metric is a finite readout of one carrier's slow response, not an extended space; on the refining source screens the rank-three band leaves bounded-energy port modes at order $2^{-l}$, and the native continuum generator has heat-trace spectral dimension two on the two-dimensional source [5, 6]. Native dependence crosses at most one seam per ordered repair event, a bound in graph hops with no laboratory speed or continuum light cone [8]. The flat $1+3$ count clock of the spacetime construction uses a specified population and read law [1]. An inflationary realization therefore requires the same events and clock to support physical expansion, stress, freezeout and scale, together with the radial, transfer and likelihood conditions above; smooth Einstein promotion requires same-family tensor-curvature convergence or the continuum small-ball/null-balance identification.

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